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# Enhancing service-learning through Generative AI: a mixed-methods study on educational Game design in a finance course

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**ABSTRACT**

Since the release of ChatGPT in late 2022, the integration of Generative Artificial Intelligence (GenAI) in higher education has posed both opportunities and challenges, especially around critical thinking and assessment integrity. This study explores how GenAI can be ethically and effectively integrated into a service-learning course on personal finance at a Hong Kong university. Students in the GenAI-supported class designed educational games for elderly participants in community centres, applying financial concepts through AI-assisted content generation and game development. Adopting a sequential explanatory mixed-methods design, we compared two student groups (GenAI-supported vs. non-GenAI) using a modified Service-Learning Outcomes Measurement Scale (S-LOMS). Quantitative results revealed significantly higher gains in Knowledge Application, Critical Thinking, and Self-Efficacy among the GenAI group. Qualitative thematic analysis of student reflections supported these findings and highlighted four key themes: enhanced knowledge transfer, increased self-efficacy, critical engagement with AI, and ethical awareness. Contrary to concerns about over-reliance, students used GenAI critically, supplementing and challenging AI outputs. These results suggest that GenAI, when scaffolded by ethical and reflective pedagogies, can enhance cognitive and metacognitive outcomes in service-learning. Implications for integrating GenAI into experiential learning and practical guidelines for ethical use are discussed.

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**SUBJECTS**
Social Sciences; Education; Higher Education; Social Sciences; Education; Higher Education; Study of Higher Education; Social Sciences; Education; Higher Education; Teaching & Learning

## 1. Introduction

### *1.2. Generative Artificial Intelligence and its impact on higher education*

The release of Chat Generative Pre-Trained Transformer (ChatGPT) in late 2022 marked a turning point in higher education. Generative Artificial Intelligence (GenAI) tools, such as ChatGPT, have shown tremendous potential to support teaching, learning, and assessment by providing immediate feedback, aiding in writing and research, and enhancing productivity (OpenAI, 2023). However, their rapid adoption has also raised critical pedagogical concerns. Educators worry that students may become overly reliant on AI-generated content (Adel et al., 2024), leading to reduced critical thinking, surface learning, and ethical blind spots (Bin-Nashwan et al., 2023; Cotton et al., 2024; Smolansky et al., 2023). Institutions have responded inconsistently, often lacking clear frameworks for responsible integration (Korseberg & Elken, 2025).

These challenges are particularly pressing within the broader shift toward Education 4.0, which emphasizes 21st-century competencies such as digital literacy (Kong et al., 2021; Casal-Otero et al., 2023), creative problem-solving, and ethical reasoning (Kapasheva et al., 2024; Peláez-Sánchez et al.,

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2024; Salmon, 2019). As such, educators are under increasing pressure to rethink instructional design and assessment practices that align with these goals while mitigating the risks of uncritical GenAI use. One emerging response is the integration of GenAI within experiential learning models—approaches that prioritise hands-on, reflective, and real-world learning experiences (Kolb, 1984). These models offer opportunities for students not only to apply knowledge but also to engage metacognitively with how that knowledge is constructed and communicated.

Among experiential pedagogies, service-learning (S-L) stands out for its ability to foster civic responsibility, personal growth, and authentic application of academic concepts (Bringle & Clayton, 2023). Recent work has begun to explore how S-L can be enhanced by digital technologies (Ngai et al., 2024; Salam et al., 2019), yet few studies have examined how GenAI might interact with the reflective and ethical components central to S-L. Concerns remain that GenAI might short-circuit the very reflective and critical processes that S-L seeks to cultivate.

This study responds to these challenges by investigating the integration of GenAI tools into a service-learning finance course at a Hong Kong university. Students designed educational games for elderly participants, using GenAI to support content creation and idea development. The course provided a unique opportunity to observe how GenAI affects students' learning outcomes and ethical awareness in a structured, community-based setting.

Using a sequential explanatory mixed-methods design, this study compares student outcomes between GenAI-supported and non-GenAI classes. It further explores how students described their use of GenAI through reflective writing.

## 2. Literature review

### 2.1. Challenges of GenAI in higher education

Since the launch of ChatGPT, universities have wrestled with how to balance the benefits and risks of GenAI in academic settings. On one hand, GenAI can facilitate personalised learning, provide instant feedback, and support creativity in writing and problem-solving (Johri et al., 2023). On the other, it may erode students' critical thinking and reflection if used uncritically, leading to what scholars have termed 'metacognitive laziness' (Fan et al., 2025; Smolansky et al., 2023). Institutions initially responded with caution or paralysis, as they struggled to understand the pedagogical implications of tools that can generate coherent, seemingly credible content with minimal effort (Korseberg & Elken, 2025). These tensions are particularly visible in assessment, where traditional models are increasingly challenged by AI's ability to simulate student work.

These issues highlight a broader tension: while GenAI can increase efficiency, it may also compromise the deeper learning processes educators aim to cultivate. Addressing this requires thoughtful integration of GenAI into pedagogies that emphasise process over product, encouraging students to reflect, critique, and apply knowledge rather than merely reproduce it.

### 2.2. Education 4.0 and the assessment paradigm shift

The concerns raised by GenAI resonate with the goals of Education 4.0, which aims to equip students with the competencies needed for a rapidly evolving, technology-rich society. These include not only digital literacy and adaptability, but also self-direction, critical thinking, and ethical reasoning (Peláez-Sánchez et al., 2024; Salmon, 2019). Education 4.0 also advocates for authentic, formative assessments that go beyond rote learning (Bennett, 2011).

Technology-enhanced assessments—such as real-time feedback, adaptive tasks, and portfolio-based evaluations—offer promising avenues in this regard. However, the emergence of GenAI adds complexity: AI tools can support process-based learning but also challenge the authenticity of assessment unless scaffolded appropriately (Chan & Colloton, 2024). This shift calls for pedagogical models that not only use GenAI tools but also guide students in evaluating and ethically using them.

### 2.3. Service-learning as a resilient pedagogy

Within this evolving landscape, service-learning (S-L) provides a compelling framework for integrating GenAI. As an experiential pedagogy, S-L blends academic learning with community service to foster personal development, civic responsibility, and applied knowledge (Bringle & Clayton, 2023). Core components include meaningful community service, integration of academic content, guided reflection, and strong community partnerships (Wang et al., 2022).

Technology-assisted S-L has become increasingly common, especially in the wake of the COVID-19 pandemic (Khiatani et al., 2023; Ngai et al., 2024; Salam et al., 2019; Xiao et al., 2022). Tools such as digital collaboration platforms and online reflection journals have expanded the possibilities for participation and engagement. However, challenges persist, including uneven engagement, lack of personalisation, and difficulty sustaining reflective depth (Taylor et al., 2019). The use of GenAI in S-L remains underexplored, particularly in how it might support or undermine core values such as ethical reasoning, empathy, and critical reflection.

### 2.4. Integrating GenAI into experiential learning

Recent studies suggest that GenAI may complement experiential learning by supporting authentic assessment, personalised learning, and project-based applications (Cacicio & Riggs, 2023; Salinas-Navarro et al., 2024a). In particular, GenAI can assist students in generating content, exploring diverse ideas, and receiving feedback during the iterative design of real-world tasks. Hsu et al. (2022) found that combining GenAI with experiential learning improved computational thinking and engagement among secondary students. This trajectory is echoed by Samala and Rawas (2025), who highlight how the integration of AI with other technologies—such as Augmented Reality (AR) and Natural Language Processing (NLP)—can foster immersive, conversational, and experiential learning environments that deepen student engagement.

However, integration must be done thoughtfully. As Goldberg and Robson (2023) argue, GenAI works best when embedded in guided experiential frameworks that include human oversight and reflection. A recent meta-analytic evidence also supports this view: Wang and Fan (2025) found that ChatGPT has a moderate effect on higher-order thinking, but only when appropriate scaffolding is present, such as when instructors position ChatGPT as a learning partner. Their study also found a large effect on learning performance, particularly when structured frameworks like Bloom's Taxonomy are used to guide AI-supported tasks. The alignment between GenAI and Kolb's Experiential Learning Cycle (1984)—comprising concrete experience, reflective observation, abstract conceptualisation, and active experimentation—is promising. GenAI can support each stage, but only if students remain active agents rather than passive recipients of AI outputs.

### 2.5. Conceptual frameworks and theoretical gaps

Several frameworks support the integration of GenAI and experiential learning. Salam et al. (2019) proposed a technology-assisted S-L model grounded in experiential principles. Salinas-Navarro et al. (2024b) developed a GenAI-aligned experiential learning framework highlighting interactive and authentic learning environments. Similarly, Ng et al. (2021) introduced an AI Literacy Framework based on Bloom's Taxonomy, which emphasises higher-order thinking and ethical AI use.

These frameworks converge on key principles: the need for active learning, critical engagement, and ethical reflection. Yet empirical studies examining their application in GenAI-enhanced S-L contexts remain scarce. Few studies combine both quantitative and qualitative methods to evaluate how GenAI shapes learning outcomes, ethical awareness, and reflective practice. This study addresses that gap.

### 2.6. Purpose of the study

This research investigates how GenAI can be integrated into a service-learning finance course, where students used AI to support the design of educational games for elderly participants. The study adopts a

**sequential explanatory mixed methods design** to assess and explain the impact of GenAI on student learning outcomes and ethical awareness.

The research questions are:

1. To what extent can GenAI improve students' service-learning outcomes and engagement compared to non-GenAI classes?
2. How does GenAI contribute to students' learning and ethical development during service-learning?

By combining survey and reflection data, this study aims to deepen understanding of how GenAI tools influence experiential learning and provide practical insights for ethical and effective integration.

## 3. Method

### 3.1. About the course

This study was conducted in the undergraduate course 'Personal Finance and Society,' offered at a private university in Hong Kong. The 14-week course covers foundational financial knowledge—including budgeting, savings, retirement planning, and scam prevention—and incorporates a service-learning (S-L) component. Students worked in groups to design educational games for elderly community centre participants, aiming to translate complex financial concepts into accessible and engaging learning tools. The S-L activities were designed to promote applied learning, civic responsibility, and personal development.

In Semester 2 (GenAI class), students were introduced to Generative AI tools and attended a Game-Based Learning workshop. In contrast, the Semester 1 (non-GenAI class) followed the same syllabus and project format but did not include GenAI or the workshop component. This distinction forms the basis of the comparative research design. Figure 1 illustrated students' works and the S-L activities after they applied GenAI tools to design games.


Figure 1. S-L activity with elderly participants (faces blurred for privacy).

### 3.2. Game-based learning and GenAI workshop

A two-hour workshop was delivered in Week 7 of the GenAI class, focusing on foundational game design and ethical GenAI use in education. Facilitated by an instructional designer, the workshop aimed to help students:

- Apply game design principles to create educational experiences.
- Articulate game ideas aligned with financial learning objectives.
- Integrate course content into structured game elements.
- Explore how GenAI (ChatGPT 3.5) could support the game creation process.

The workshop was divided into two parts. The first introduced foundational concepts of educational game design, drawing on Boller and Kapp (2017) taxonomy. Through hands-on exercises, students learned to distinguish between commercial and educational games and to map learning goals onto gameplay mechanics.

The second section focused on leveraging ChatGPT 3.5 via the university's in-house version. Students were introduced to prompt engineering techniques, including the 'write before write' method, and practiced using ChatGPT to generate content, simplify concepts, and iterate game ideas. Emphasis was placed on verifying AI-generated content and maintaining human oversight to support ethical use.

### 3.3. Research design and participants

The study adopted a sequential explanatory mixed-methods design (Creswell, 2015), involving a quantitative phase followed by a qualitative phase to explain the observed results (See Figure 2). The temporal sequence was as follows: (1) pre-test survey at the start of the course; (2) intervention (GenAI use + workshop in Semester 2 only); (3) post-test survey at course completion; (4) individual reflective essays collected immediately after the S-L activities. The quantitative data provided baseline and outcome comparisons, while the qualitative reflections offered deeper insight into students' experiences with GenAI.

A total of 77 students from the two course offerings in Academic Year 2023/24 participated. The GenAI group (n = 36) and the non-GenAI group (n = 41) were comparable in gender, year of study, and disciplinary background (see Table 1). A convenience sampling method was used. Participation was voluntary and anonymous, and ethics clearance was granted by the university's Teaching and Learning Unit, which oversees educational studies.

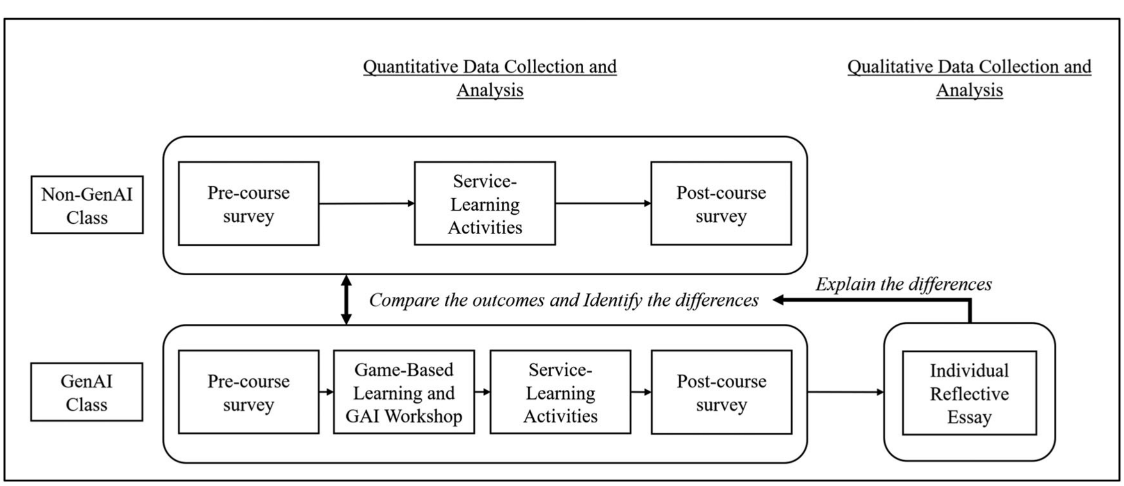


Figure 2. Research design developed by authors.

Table 1. Demographic information of the surveyed students.

| | Non-GenAI class (N = 42) | GenAI class (N = 35) |
|---|---|---|
| **Gender** | | |
| Male | 24 (57.1%) | 20 (57.1%) |
| Female | 18 (42.9%) | 15 (42.9%) |
| **Year of Study** | | |
| Year 1 | 0 (0.0%) | 1 (2.9%) |
| Year 2 | 7 (16.7%) | 14 (40.0%) |
| Year 3 | 26 (61.9%) | 13 (37.1%) |
| Year 4 | 8 (19.0%) | 7 (20.0%) |
| Year 5 or above | 1 (2.4%) | 0 (0.0%) |
| **Discipline** | | |
| Humanities and Social Sciences | 3 (7.1%) | 3 (8.6%) |
| Business | 20 (47.6%) | 15 (42.9%) |
| Science and Technology | 19 (45.2%) | 17 (48.6%) |

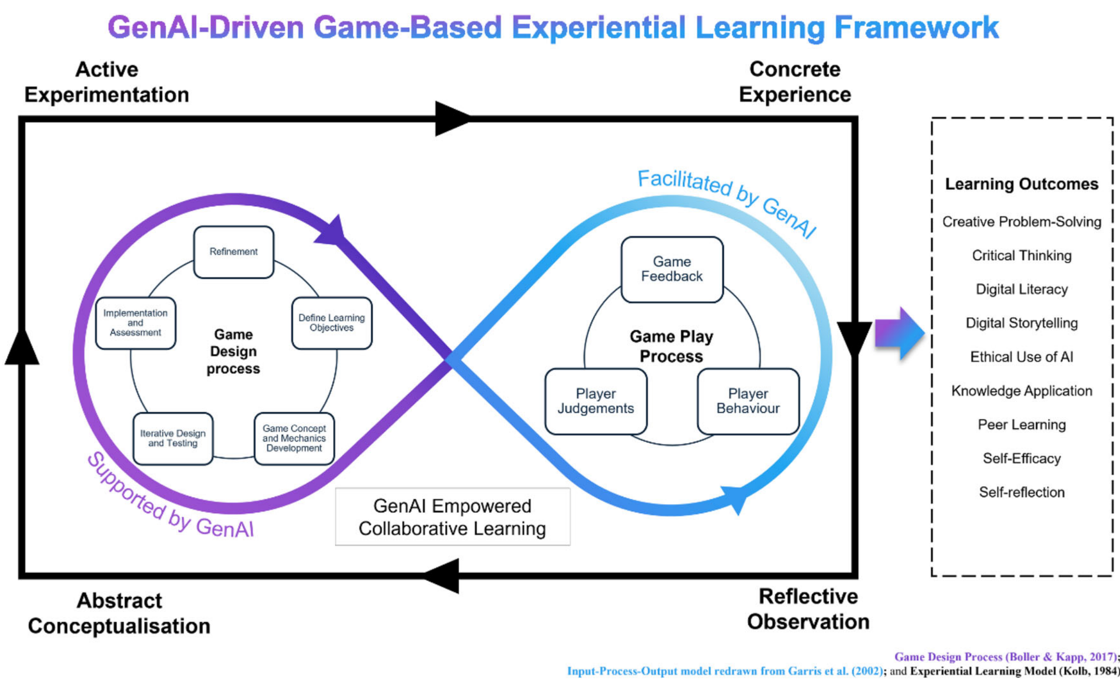


Figure 3. GenAI-Driven Game-Based Experiential Learning Framework.

### 3.4. Conceptual framework

This study adopts the GenAI-Driven Game-Based Experiential Learning Framework (see Figure 3), developed by the authors to integrate GenAI into experiential and game-based learning environments. The framework draws on three foundational models:

- Kolb's Experiential Learning Cycle (1984), which emphasizes the iterative learning stages of Concrete Experience, Reflective Observation, Abstract Conceptualization, and Active Experimentation.
- Boller and Kapp's Educational Game Design Process (2017), which outlines best practices in learning game development.
- Garris et al.'s Input-Process-Outcome Game Model (Garris et al., 2002), which structures gameplay analysis in educational contexts.

As shown in Figure 3, the framework is visualised as a double-loop model, mapping the Game Design Process onto the left loop (supported by GenAI) and the Game Play Process onto the right loop (facilitated by GenAI). These two loops are embedded within Kolb's experiential cycle, illustrating how students iteratively design and refine educational games through AI-empowered collaboration and player feedback.

The Game Design Process begins with defining learning objectives and proceeds through game concept development, iterative testing, and refinement. GenAI assists students in brainstorming ideas, simplifying complex content, and generating draft assets or texts. This process supports Kolb's Abstract Conceptualisation and Active Experimentation phases.

The Game Play Process, influenced by the Input-Process-Outcome model, involves players interacting with the games, receiving feedback, and engaging in judgment and reflection. GenAI can support this phase by generating dynamic in-game prompts or offering feedback based on player behaviour. This aligns with Concrete Experience and Reflective Observation in Kolb's model.

The framework identifies key learning outcomes expected from this integration, including: creative problem-solving, critical thinking, digital literacy, digital storytelling, ethical AI use, knowledge application, peer learning, self-efficacy, and self-reflection. These outcomes reflect the 21st-century competencies prioritised in Education 4.0.

This conceptual framework served not only to guide the intervention design (workshop and GenAI use) but also informed the thematic coding structure during qualitative analysis, ensuring alignment between theory, pedagogy, and data interpretation.

### 3.5. Quantitative approach

To measure the effectiveness of S-L in student development, the S-L Outcomes Measurement Scale (S-LOMS) (Lau & Snell, 2020) was utilized. It aims to measure and assess the developmental changes in students' learning outcomes resulting from S-L experiences and serves as a reference for S-L practitioners and researchers when selecting categories and domains for their own investigations (Lau & Snell, 2020). The categories and domains measurement was shown in **Table A-1 of** Appendix A. There are 11 domains measure students' learning in terms of 'Knowledge Application', 'Creative Problem-solving Skills', 'Relationship & Team Skills', 'Self-reflection Skills', 'Critical Thinking Skills', 'Community & Understanding', 'Caring & Respect', 'Sense of Social Responsibility', 'Self-efficacy', 'Self-understanding', and 'Commitment to Self-improvement'.

These domains under four overarching categories which is named, 'Knowledge Application', 'Personal and Professional skills', 'Civic Orientation and Engagement', and 'Self-awareness'. Participants responded on a 10-point Likert scale ranging from 1 (lowest) to 10 (highest), with higher scores indicating greater competencies. The scale demonstrated excellent internal consistency in the current study, with Cronbach's $\alpha$ of 0.96 (data not shown). Analyses were conducted using IBM SPSS 28 software.

Students' self-reported scores in each of the 11 domains were evaluated both pre- and post-course. Paired-sample t-tests were performed to examine the overall effectiveness of the GenAI and Non-GenAI courses. Preliminary results from an independent t-test showed no significant difference between the pre-test scores of GenAI and non-GenAI classes. A subsequent independent t-test determined statistically significant differences in S-L outcomes across the 11 domains between GenAI and Non-GenAI classes. The t-test findings, along with Cohen's *d* effect sizes, identified the domains where students showed enhancement with GenAI adoption. These results guided the researcher in obtaining students' reflections for further explanation of the findings.

### 3.6. Qualitative approach

To deepen the understanding of the quantitative findings, a qualitative case study approach was adopted. Students from the GenAI class submitted individual reflective essays describing how they engaged with GenAI tools during the development of their educational games. These reflections offered insight into students' experiences of applying course knowledge with AI assistance, their perceptions of GenAI's affordances and limitations, and any ethical considerations they encountered.

A thematic analysis was conducted using Braun and Clarke (2006) six-step approach: familiarisation, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the report. Two researchers independently coded the data using a manual, spreadsheet-based process. An inductive approach was used, allowing themes to emerge from the data. Discrepancies were discussed and resolved through consensus, and the resulting themes were refined in alignment with the study's conceptual framework.

The qualitative phase was informed by the quantitative results. For instance, domains showing significant improvement—such as Knowledge Application, Critical Thinking, and Self-Efficacy—became focal points for thematic interpretation. This integration of data followed the logic of the sequential explanatory mixed-methods design, where qualitative insights explain and contextualise the statistical trends. Together, this process enabled triangulation and strengthened the overall trustworthiness of the findings.

### 3.7. Ethical considerations

This study was conducted in accordance with the university's institutional ethical guidelines, and ethics clearance was granted by the university's Teaching and Learning Unit, which oversees educational studies. All participants were fully informed about the purpose of the study, and written informed consent was obtained prior to participation. Participants were assured of confidentiality, anonymity, and their right to withdraw at any time without consequence.

## 4. Findings

### 4.1. Preliminary analysis

Preliminary analysis revealed no significant differences in the 11-domain S-L outcomes based on gender or year of study except Caring & Respect in Pre-test. Female participants indicated a higher agreement on this domain (data not shown). However, this gender difference disappeared at post-test, suggesting a potential levelling effect from the course experience. Levene's tests confirmed homogeneity of variances for t-tests. There were no missing data in this study.

### 4.2. Paired-sample t-test of the 11 domains

Figure 4 and Table 2 present the results of the paired-sample t-tests. Significant improvements were observed in both classes. However, the paired-sample t-test results indicate that GenAI classes exhibited greater improvements in most of the S-LOMS areas compared to non-GenAI classes. The effect sizes

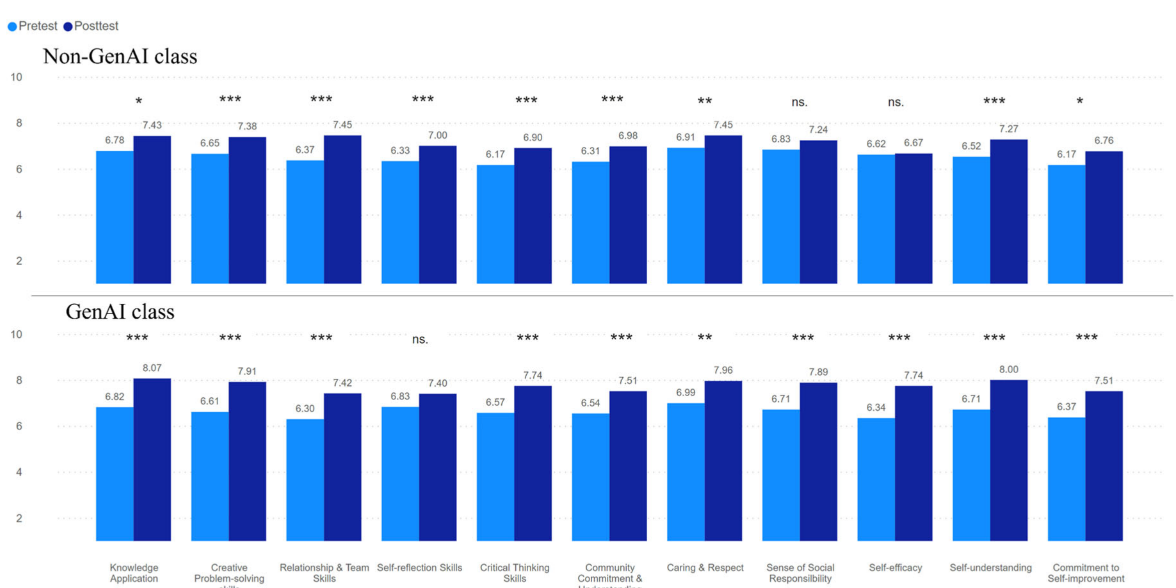


**Figure 4.** Bar chart Presentation of the Comparison of Pre and Post-test scores of the S-LOM 11 domains: Non-GenAI (Semester 1) vs. GenAI Classes (Semester 2).
*Note:* $p < .05^{*}$; $p < .01^{**}$; $p < .001^{***}$; ns. – not significant.

Table 2. Comparison of pre and post-test scores of the S-LOM 11 domains: Non-GenAI (Semester 1) vs. GenAI classes (Semester 2) - Paired samples t-test.

| Domains | Pre-test mean | Post-test mean | Mean differences (SD) | SEM | 95% CI lower | 95% CI upper | *t* | *df* | *p* | *d* |
|---|---|---|---|---|---|---|---|---|---|---|
| **Non-GenAI Class** | | | | | | | | | | |
| Knowledge Application | 6.78 | 7.43 | 0.65 (1.75) | 0.27 | 0.10 | 1.20 | 2.41 | 41 | 0.02 | 0.37 |
| Creative Problem-solving skills | 6.65 | 7.38 | 0.73 (1.53) | 0.24 | 0.25 | 1.21 | 3.09 | 41 | <0.001 | 0.48 |
| Relationship & Team Skills | 6.37 | 7.45 | 1.09 (1.24) | 0.19 | 0.70 | 1.47 | 5.68 | 41 | <0.001 | 0.88 |
| Self-reflection Skills | 6.33 | 7.00 | 0.67 (1.43) | 0.22 | 0.22 | 1.11 | 3.03 | 41 | <0.001 | 0.47 |
| Critical Thinking Skills | 6.17 | 6.90 | 0.74 (1.25) | 0.19 | 0.35 | 1.13 | 3.83 | 41 | <0.001 | 0.59 |
| Community Commitment & Understanding | 6.31 | 6.98 | 0.67 (1.26) | 0.19 | 0.27 | 1.06 | 3.42 | 41 | <0.001 | 0.53 |
| Caring & Respect | 6.91 | 7.45 | 0.54 (1.30) | 0.20 | 0.14 | 0.94 | 2.70 | 41 | 0.01 | 0.42 |
| Sense of Social Responsibility | 6.83 | 7.24 | 0.40 (1.50) | 0.23 | −0.06 | 0.87 | 1.75 | 41 | 0.09 | 0.27 |
| Self-efficacy | 6.62 | 6.67 | 0.05 (1.36) | 0.21 | −0.38 | 0.47 | 0.23 | 41 | 0.82 | 0.03 |
| Self-understanding | 6.52 | 7.27 | 0.75 (1.23) | 0.19 | 0.37 | 1.13 | 3.95 | 41 | <0.001 | 0.61 |
| Commitment to Self-improvement | 6.17 | 6.76 | 0.60 (1.58) | 0.24 | 0.10 | 1.09 | 2.44 | 41 | 0.02 | 0.38 |
| **GenAI Class** | | | | | | | | | | |
| Knowledge Application | 6.82 | 8.07 | 1.25 (1.78) | 0.30 | 0.64 | 1.86 | 4.16 | 34 | <0.001 | 0.70 |
| Creative Problem-solving skills | 6.61 | 7.91 | 1.30 (2.06) | 0.35 | 0.60 | 2.01 | 3.75 | 34 | <0.001 | 0.63 |
| Relationship & Team Skills | 6.30 | 7.42 | 1.12 (2.06) | 0.35 | 0.42 | 1.83 | 3.22 | 34 | <0.001 | 0.54 |
| Self-reflection Skills | 6.83 | 7.40 | 0.57 (2.40) | 0.41 | −0.25 | 1.40 | 1.41 | 34 | 0.17 | 0.24 |
| Critical Thinking Skills | 6.57 | 7.74 | 1.17 (2.09) | 0.35 | 0.45 | 1.89 | 3.31 | 34 | <0.001 | 0.56 |
| Community Commitment & Understanding | 6.54 | 7.51 | 0.97 (1.85) | 0.31 | 0.34 | 1.61 | 3.11 | 34 | <0.001 | 0.53 |
| Caring & Respect | 6.99 | 7.96 | 0.97 (1.94) | 0.33 | 0.30 | 1.64 | 2.96 | 34 | 0.01 | 0.50 |
| Sense of Social Responsibility | 6.71 | 7.89 | 1.17 (1.98) | 0.33 | 0.49 | 1.85 | 3.50 | 34 | <0.001 | 0.59 |
| Self-efficacy | 6.34 | 7.74 | 1.40 (2.23) | 0.38 | 0.64 | 2.16 | 3.72 | 34 | <0.001 | 0.63 |
| Self-understanding | 6.71 | 8.00 | 1.29 (1.78) | 0.30 | 0.67 | 1.90 | 4.27 | 34 | <0.001 | 0.72 |
| Commitment to Self-improvement | 6.37 | 7.51 | 1.14 (2.14) | 0.36 | 0.41 | 1.88 | 3.15 | 34 | <0.001 | 0.53 |

*Note:* Equal Variance Assumed and Two-tailed Test.

(Cohen's *d*) further highlight the magnitude of these improvements. Notably, three pairs demonstrated significant differences.

#### 4.2.1. Knowledge application

For the first pair, the GenAI class exhibited a significantly higher mean difference (M = 1.25, SD = 1.78) compared to the non-GenAI class (M = 0.65, SD = 1.75). A statistically significant improvement for the GenAI class, $t$ (34) = 4.16, $p < .001$. The effect size for GenAI Class was $d = 0.70$, compared to 0.37 in non-GenAI class, indicating a large effect in GenAI class.

#### 4.2.2. Self-efficacy

A notable improvement was observed in the domain of Self-efficacy, where the GenAI class showed a mean difference of 1.40 (SD = 2.23), compared to the non-GenAI class with a mean difference of only 0.05 (SD = 1.36). The t-test results were significant, $t$ (34) = 3.72, $p < .001$. The effect size for GenAI class was $d = 0.63$, compared to 0.04 in non-GenAI class, indicating a medium to large effect in GenAI classes.

### 4.3. Paired-sample t-test of the 4 overarching categories

In the other hand, Table 3 presents the paired-sample t-test results comparing the pre- and post-test scores of the S-LOMS categories between the non-GenAI class and the GenAI class. Overall, both classes showed significant improvements across all domains. However, the GenAI class exhibited greater improvements in each domain compared to the non-GenAI class. Those differences could be observed via their effect size.

### 4.4. Independent sample t-test of the 11 domains and 4 overarching categories for comparing post-test S-LOM scores between non-GenAI and GenAI classes

As mentioned before, no difference was found between the pre-test scores of GenAI and non-GenAI classes. The independent samples t-test results comparing post-test S-LOM scores between non-GenAI and GenAI classes reveal significant differences in several domains (See Table 4 and Table 5). The

**Table 3.** Comparison of pre- and post-test scores of the S-LOM 4 categories: Non-GenAI (Semester 1) vs. GenAI classes (Semester 2) - Paired samples t-test.

| Domains | Pre-test mean | Post-test mean | Mean differences (SD) | SEM | 95% CI lower | 95% CI upper | *t* | *df* | *p* | *d* |
|---|---|---|---|---|---|---|---|---|---|---|
| **Non-GenAI Class** | | | | | | | | | | |
| Knowledge Application | 6.78 | 7.43 | 0.65 (1.75) | 0.27 | 0.10 | 1.20 | 2.41 | 41 | 0.02 | 0.37 |
| Personal & Professional Skills | 6.44 | 7.30 | 0.86 (1.10) | 0.17 | 0.51 | 1.20 | 5.05 | 41 | <0.001 | 0.78 |
| Civic Orientation & Engagement | 6.70 | 7.26 | 0.56 (1.08) | 0.17 | 0.22 | 0.89 | 3.37 | 41 | <0.001 | 0.52 |
| Self-awareness | 6.46 | 6.99 | 0.54 (1.14) | 0.18 | 0.18 | 0.89 | 3.06 | 41 | <0.001 | 0.47 |
| S-LOMS (Overall) | 6.57 | 7.25 | 0.68 (0.99) | 0.15 | 0.37 | 0.99 | 4.48 | 41 | <0.001 | 0.69 |
| **GenAI Class** | | | | | | | | | | |
| Knowledge Application | 6.82 | 8.07 | 1.25 (1.78) | 0.30 | 0.64 | 1.86 | 4.16 | 34 | <0.001 | 0.70 |
| Personal & Professional Skills | 6.51 | 7.64 | 1.13 (1.93) | 0.33 | 0.46 | 1.79 | 3.45 | 34 | <0.001 | 0.58 |
| Civic Orientation & Engagement | 6.80 | 7.80 | 1.00 (1.78) | 0.30 | 0.39 | 1.62 | 3.34 | 34 | <0.001 | 0.56 |
| Self-awareness | 6.54 | 7.81 | 1.28 (1.77) | 0.30 | 0.67 | 1.89 | 4.28 | 34 | <0.001 | 0.72 |
| S-LOMS (Overall) | 6.64 | 7.78 | 1.14 (1.69) | 0.29 | 0.56 | 1.72 | 3.99 | 34 | <0.001 | 0.67 |

*Note:* Equal Variance Assumed and Two-tailed Test.

**Table 4.** Comparison of post-test S-LOM scores of the 11 domains between non-GenAI (Semester 1) vs. GenAI classes (Semester 2) - independent samples t-test.

| Domains | Non-GenAI class | GenAI class | Mean differences (SD) | 95% CI lower | 95% CI upper | *t* | *df* | *p* | *d* |
|---|---|---|---|---|---|---|---|---|---|
| Knowledge Application | 7.43 | 8.07 | 0.64 (0.30) | 1.24 | 0.03 | 2.10 | 75 | 0.04 | 0.48 |
| Creative Problem-solving skills | 7.38 | 7.91 | 0.53 (0.33) | 1.19 | 0.12 | 1.62 | 75 | 0.11 | 0.37 |
| Relationship & Team Skills | 7.45 | 7.42 | −0.03 (0.36) | 0.69 | 0.76 | 0.09 | 75 | 0.93 | 0.02 |
| Self-reflection Skills | 7.00 | 7.40 | 0.40 (0.44) | 1.27 | 0.47 | 0.91 | 75 | 0.36 | 0.21 |
| Critical Thinking Skills | 6.90 | 7.74 | 0.84 (0.39) | 1.61 | 0.07 | 2.17 | 75 | 0.03 | 0.50 |
| Community Commitment & Understanding | 6.98 | 7.51 | 0.53 (0.38) | 1.30 | 0.22 | 1.41 | 75 | 0.16 | 0.32 |
| Caring & Respect | 7.45 | 7.96 | 0.51 (0.35) | 1.21 | 0.19 | 1.46 | 75 | 0.15 | 0.33 |
| Sense of Social Responsibility | 7.24 | 7.89 | 0.65 (0.35) | 1.35 | 0.06 | 1.83 | 75 | 0.07 | 0.42 |
| Self-efficacy | 6.67 | 7.74 | 1.07 (0.45) | 1.98 | 0.17 | 2.37 | 75 | 0.02 | 0.54 |
| Self-understanding | 7.27 | 8.00 | 0.73 (0.37) | 1.46 | 0.00 | 1.98 | 75 | 0.05 | 0.45 |
| Commitment to Self-improvement | 6.76 | 7.51 | 0.75 (0.43) | 1.61 | 0.10 | 1.76 | 75 | 0.08 | 0.40 |
| S-LOMS (Overall) | 7.25 | 7.78 | 1.25 (0.30) | 1.12 | 0.06 | −1.79 | 75 | 0.08 | 0.41 |

*Note:* Equal Variance Assumed and Two-tailed Test.

**Table 5.** Comparison of post-test S-LOM scores of the 4 categories between non-GenAI (Semester 1) vs. GenAI classes (Semester 2) - independent samples t-test.

| Domains | Non-GenAI class | GenAI class | Mean differences (SD) | 95% CI lower | 95% CI upper | *t* | *df* | *p* | *d* |
|---|---|---|---|---|---|---|---|---|---|
| Knowledge Application | 7.43 | 8.07 | 0.64 (0.30) | 1.24 | 0.03 | 2.10 | 75 | 0.04 | 0.48 |
| Personal & Professional Skills | 7.30 | 7.64 | 0.34 (0.32) | 0.99 | 0.30 | 1.06 | 75 | 0.29 | 0.24 |
| Civic Orientation & Engagement | 7.26 | 7.80 | 0.54 (0.32) | 1.19 | 0.11 | 1.67 | 75 | 0.10 | 0.38 |
| Self-awareness | 6.99 | 7.81 | 0.82 (0.36) | 1.54 | 0.10 | 2.28 | 75 | 0.03 | 0.52 |
| S-LOMS (Overall) | 7.25 | 7.78 | 1.25 (0.30) | 1.12 | 0.06 | 1.79 | 75 | 0.08 | 0.41 |

*Note:* Equal Variance Assumed and Two-tailed Test.

GenAI class showed higher mean scores in Knowledge Application ($M = 8.07$) compared to the non-GenAI class ($M = 7.43$), $t$ (75) $= 2.10$, $p = .04$. Critical Thinking Skills also significantly higher in the GenAI class, $t$ (75) $= 2.17$, $p = .03$. A moderate effect size ($d = 0.50$) was reported. Similar to the pair-sample t-test results, self-efficacy showed a substantial increase in the GenAI class ($M = 7.74$) compared to the non-GenAI class ($M = 6.67$), with a mean difference of 1.07, $t$ (75) $= 2.37$, $p = 0.02$, and a moderate effect size ($d = 0.54$). The GenAI class students indicated a higher post-test scores when comparing to non-GenAI students.

### 4.5. Thematic analysis

To explain the quantitative findings, a thematic analysis was conducted on student reflections from the GenAI class. The six-step process (Braun & Clarke, 2006) and was guided by both inductive coding and deductive alignment with the conceptual framework (Kolb's Experiential Learning Cycle and the GenAI-Driven Game-Based Experiential Learning Framework). Four key themes were identified, corresponding to the domains in which GenAI students showed the most significant improvements:

*Knowledge Application*, *Critical Thinking*, and *Self-Efficacy*. A fourth theme concerning *Ethical Awareness* also emerged, illustrating how students engaged with the boundaries and responsibilities of AI use.

#### *4.5.1. GenAI as a knowledge Application catalyst*

Students reported that GenAI, particularly ChatGPT, helped them apply course concepts more effectively in real-world contexts. It simplified technical language, supported idea generation, and provided structured outlines for their game designs. As one student noted:

> ChatGPT's ability to organize financial information efficiently was invaluable in creating clear and concise leaflets about Silver Bonds.

Another described how AI supported fraud education content:

> We used ChatGPT to reference various scams, such as AI scams, online scams, etc. After modification, the questions were designed to help participants and us understand different types of fraud and how to deal with it.

These examples reflect Kolb's Concrete Experience and Abstract Conceptualisation stages, where learners engage directly with authentic tasks and integrate theory with practice. GenAI acted as a bridge between classroom knowledge and the needs of elderly learners, enabling students to better contextualise financial concepts in accessible ways.

#### *4.5.2. Boosting self-efficacy and creative problem-solving*

Students frequently described how GenAI boosted their confidence and supported creativity in project development. This theme was particularly prominent among those who initially felt unsure about designing games or explaining finance to non-experts. One student wrote:

> Combining our own ideas with those from ChatGPT, we developed a role-playing game that used the government's 'Money Tracker' app.

Others highlighted the role of AI in refining content under time pressure:

> With limited space available, GenAI's ability to streamline information and focus on key points was extremely useful.

These experiences correspond to Active Experimentation in Kolb's cycle, as students used GenAI to test, revise, and implement their designs iteratively. GenAI was viewed not as a crutch, but as a collaborative tool for enhancing their output.

#### *4.5.3. Fostering critical thinking through AI evaluation*

While students appreciated GenAI's utility, many were critical of its limitations. Some encountered outdated or incorrect information and described how this prompted deeper analysis and fact-checking. For example:

> When I used ChatGPT to answer questions, I found that it completely distorted the concept of bank bonds, which made me realize that I relied too much on ChatGPT and lost my own subjective initiative.

Another student echoed this skepticism:

> We realized the information which ChatGPT provided was different from our own research. After cross-checking, we figured out the information was outdated.

These moments illustrate Reflective Observation, where students moved from passive consumption to critical engagement. Rather than undermining learning, GenAI triggered metacognitive questioning—an essential aspect of higher-order thinking. These findings align with recent research emphasising that GenAI fosters critical thinking *when students are guided to question it* (Fan et al., 2025; Wang & Fan, 2025).

#### *4.5.4. Ethical awareness and responsible AI use*

Students demonstrated thoughtful consideration of the ethical implications of GenAI use. Rather than uncritically accepting its output, they reflected on when and how AI should be used responsibly. One student noted:

> While (G)AI tools can offer valuable support in generating ideas and questions, their biggest advantage lies in reorganizing information and providing simpler and more concise texts. It is essential to complement their capabilities with human knowledge and judgment.

Others discussed verifying AI content against official sources, such as government websites, and emphasised human oversight in interpreting outputs.

These reflections suggest an emerging form of ethical reflexivity, as students recognised the importance of accountability in using AI in real-world scenarios. This theme resonates with Kolb's Reflective Observation and Active Experimentation stages, as well as with broader calls in the literature for integrating AI literacy and ethical use into GenAI-based education (Cotton et al., 2024; Ng et al., 2021).

### *4.6. Summary of findings*

The integrated results of this mixed-methods study indicate that the GenAI-supported service-learning experience led to significantly greater improvements in student learning outcomes than the traditional, non-GenAI version of the course. Quantitative analyses showed that students in the GenAI class reported larger gains in key domains such as *Knowledge Application*, *Critical Thinking*, and *Self-Efficacy*, with moderate to large effect sizes ($d = 0.56$ to 0.70). These improvements were also reflected at the category level, suggesting that GenAI influenced not just isolated skills, but broader developmental outcomes related to personal growth and civic engagement.

Thematic analysis of student reflections provided insight into the mechanisms behind these quantitative gains. Students described how GenAI enhanced their ability to apply theoretical knowledge, supported creative and iterative problem-solving, and encouraged critical evaluation of AI-generated content. Importantly, they also reflected on the ethical responsibilities associated with AI use, highlighting a growing awareness of its limitations and potential biases.

## 5. Discussion

### *5.1. Interpretation of the findings*

This study provides evidence that integrating GenAI into a service-learning (S-L) finance course can enhance key student learning outcomes. Quantitative results indicated statistically significant improvements in *Knowledge Application*, *Critical Thinking*, and *Self-Efficacy*, with moderate to large effect sizes. Qualitative reflections supported these findings, revealing how students leveraged GenAI tools to structure content, ideate games, and critically evaluate the limitations of AI-generated information. While the sample size was small, consistent effect patterns across multiple domains lend preliminary support to the hypothesis that GenAI, when scaffolded appropriately, contributes meaningfully to both cognitive and reflective development.

However, these findings must be interpreted cautiously. With only two classes from a single institution, the generalisability is limited. The large effect sizes observed may be influenced by context-specific factors such as instructor style or student motivation. Additionally, the use of self-report data can introduce bias. Future studies should incorporate objective data sources—such as peer assessments, prompt quality ratings, or usage logs—to triangulate learning gains.

### *5.2. Connection to the previous research*

The results of this study build upon and extend existing research on the educational affordances of Generative AI (GenAI) and service-learning (S-L) pedagogy. Prior studies have suggested that GenAI can support learning by providing feedback, generating content drafts, and accelerating ideation (e.g. Cacicio & Riggs, 2023; Chan & Colloton, 2024). However, these benefits are often framed in general classroom or experimental settings (Wang & Fan, 2025). This study situates GenAI within a service-learning context, where its use is tied to a concrete societal problem and collaborative design process. The significant gains observed in *Knowledge Application* and *Critical Thinking* suggest that GenAI's potential may be most effectively realised when embedded in authentic, purpose-driven learning environments that go beyond academic performance.

This study also complements concerns raised in earlier literature about the risk of cognitive offloading and surface-level learning when students use GenAI without guidance (Fan et al., 2025; Smolansky et al., 2023). In contrast, the present findings demonstrate that when students are required to reflect on, adapt, and verify AI-generated outputs—such as tailoring financial explanations for elderly learners—they engage in metacognitive regulation and higher-order thinking. These findings supported that GenAI tools enhance higher order thinking only when scaffolding and structured frameworks are in place.

The results also reinforce and expand established theoretical frameworks in education. Specifically, students' learning trajectories followed patterns consistent with Bloom's Taxonomy (Bloom et al., 1956): starting from knowledge recall and comprehension, moving through application and analysis, and culminating in evaluation of GenAI outputs and ethical decision-making. The qualitative themes illustrate how students critically assessed AI-generated content and reflected on its limitations—indicating active engagement at multiple cognitive levels.

Simultaneously, the findings reflect Kolb's Experiential Learning Cycle (Kolb, 1984). Students described using GenAI during both the Abstract Conceptualisation and Active Experimentation stages while building their games, and engaged in Reflective Observation when evaluating AI-generated inaccuracies or reworking content for community users. This layered learning process—shaped by experimentation, iteration, and feedback—highlights how GenAI, when aligned with experiential principles, can act as a scaffold rather than a substitute for critical thinking and creativity.

Furthermore, the study addresses a key gap in the literature by demonstrating a synergistic integration of GenAI within S-L—a pairing rarely explored empirically. While S-L is traditionally rooted in social engagement and reflective practice (Bringle & Clayton, 2023), this study shows that GenAI can extend its reach by facilitating rapid prototyping, accessible content generation, and deeper ethical reflection. In this way, the research provides not just affirmation of existing theories, but a step toward a new hybrid model of AI-enhanced experiential education.

These findings not only extend existing research on the cognitive benefits of GenAI integration but also point toward a deeper pedagogical value: the opportunity to cultivate ethical judgment through structured AI use. While much of the literature focuses on skill development and performance outcomes, this study also revealed how students encountered and responded to the ethical challenges inherent in working with AI. This dimension warrants closer attention.

### 5.3. Ethical considerations

This study found that students' ethical awareness of GenAI use emerged not solely from explicit instruction, but through the design of the learning task itself – particularly its socially accountable and community-facing nature. Designing educational games for older adults created a real-world context in which the accuracy and appropriateness of AI-generated content had direct implications. This heightened sense of responsibility motivated students to verify content, refine prompts, and consult multiple sources when inaccuracies were identified.

Such behaviour reflected deeper learning aligned with Bloom's higher-order thinking stages—analysis, evaluation, and creation—as well as social responsibility. These moments also align with Kolb's Reflective Observation and Abstract Conceptualization phases, where students critically reflected on AI limitations and adjusted their outputs accordingly. In line with Fan et al. (2025), this critical engagement helped reduce overreliance on AI tools and fostered deeper metacognitive reflection.

However, not all students demonstrated the same level of discernment. A minority relied on AI outputs without sufficient scrutiny, highlighting the need for scaffolding ethical awareness through structured reflection, feedback and peer discussion. Overall, the findings suggest that when GenAI is integrated into authentic, socially accountable learning context, it can serve as a meaningful catalyst for ethical development.

## 6. Implications

This study offers several practical implications for educators, curriculum designers, and institutions integrating GenAI into experiential learning contexts such as service-learning (S-L). First, the GenAI-Driven Game-Based Experiential Learning Framework provides a replicable model for fostering creativity, critical thinking, and knowledge application. Designing educational games with GenAI encouraged students to move beyond content consumption toward active knowledge construction, particularly when scaffolded through Kolb's learning cycle.

Second, GenAI tools can serve as a bridge between digital literacy and ethical AI use. However, this study highlights that ethical competence does not emerge automatically. Educators must embed deliberate scaffolds—such as structured reflection prompts, peer feedback, and source triangulation—into AI-supported tasks to promote responsible usage.

Third, implementation must account for challenges. Students vary in GenAI fluency, and some defaulted to uncritical reliance. Training in prompt engineering, clear usage policies, and discussion of AI limitations should be embedded early in the curriculum. Institutional infrastructure—including AI usage policies, staff development opportunities, and equitable access to GenAI tools—is essential to scale such practices effectively.

When thoughtfully integrated, GenAI can enrich experiential learning—not just by enhancing academic outcomes, but by fostering metacognitive growth, ethical reasoning, and digital responsibility. These competencies are vital for preparing students to navigate a rapidly evolving technological landscape.

## 7. Limitations and future research directions

This study offers promising insights into the integration of GenAI within a service-learning (S-L) context; however, several limitations should be acknowledged. First, the relatively small sample size and single-institution scope limit the generalizability of the findings. Second, the study focused on a financial literacy course, which may not reflect outcomes in other disciplines. Third, this study relied primarily on self-reported data and student reflections, while rich in insight, are subject to potential bias. In particular, students' perceptions of skills such as critical thinking and self-efficacy may not fully align with their actual performance. Future studies could incorporate objective performance-based assessments—such as peer evaluations, behavioural analytics, or AI usage logs—to triangulate and validate self-reported outcomes.

Additionally, this study employed ChatGPT-3.5, a model with known limitations in accuracy and contextual depth compared to newer large language models (LLMs). Future studies could compare learning outcomes across different GenAI tools and investigate how various GenAI functionalities (e.g. multimodal input, real-time feedback) affect student learning.

Building on this foundation, future research should explore the long-term impact of GenAI-supported learning, particularly whether improvements in knowledge application, critical thinking, and self-efficacy are sustained over time. There is also a need to design validated frameworks for assessing students' ethical engagement with GenAI. Lastly, the role of prompt engineering—the ability to formulate effective queries—emerged as an important, yet underexplored skill. Investigating how this skill evolves and supports learning could help educators develop more targeted interventions and training strategies.

## 8. Conclusion

This study examined how integrating Generative AI (GenAI) into a service-learning (S-L) finance course can enhance students' learning outcomes. By engaging students in the design of educational games for elderly community members, the GenAI-supported curriculum enabled learners to apply theoretical financial concepts in authentic, socially meaningful contexts.

Quantitative findings revealed that students in the GenAI-integrated class demonstrated greater improvements in *Knowledge Application, Critical Thinking Skills, and Self-Efficacy* compared to those in the non-GenAI group. Thematic analysis of student reflections supported these outcomes and revealed how GenAI served not only as a knowledge and design support tool, but also as a catalyst for ethical reflection and metacognitive engagement.

Framed within Kolb's Experiential Learning Cycle and supported by the GenAI-Driven Game-Based Experiential Learning Framework, this study contributes new evidence to the field by showing that GenAI, when meaningfully integrated, can deepen experiential learning. It promotes higher-order thinking, creativity, and ethical awareness—skills essential in today's technology-driven world.

These findings underscore the potential of GenAI as a pedagogical partner rather than a substitute for human learning. When students are empowered as critical users and ethical evaluators of GenAI, the technology becomes a transformative tool for cultivating reflective, future-ready learners.

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## Data availability statement

The data supporting the findings of this study are available from the corresponding author upon request.

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